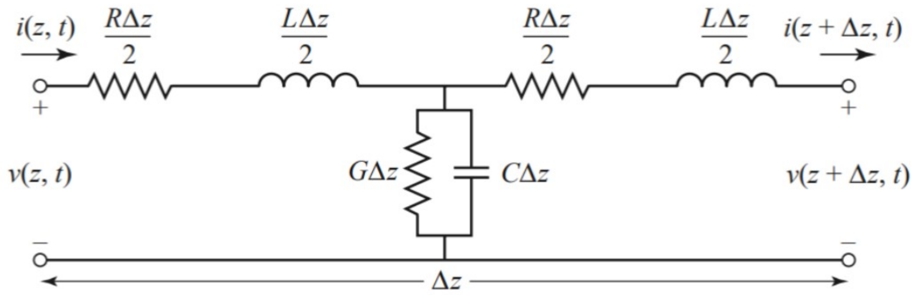
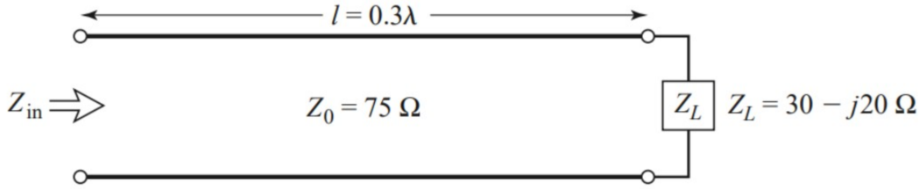
2.3 RG-402U semirigid coaxial cable has an inner conductor diameter of 0.91 mm and a dielectric diameter (equal to the inner diameter of the outer conductor) of 3.02 mm. Both conductors are copper, and the dielectric material is Teflon. Compute the R, L, G, and C parameters of this line at 1 GHz, and use these results to find the characteristic impedance and attenuation of the line at 1 GHz. Compare your results to the manufacturer's specifications of 50 Ω and 0.43 dB/m, and discuss reasons for the difference.

2.7 Show that the T-model of a transmission line shown in the accompanying figure also yields the telegrapher equations derived in Section 2.1.



2.8 A lossless transmission line of electrical length $\ell = 0.3\lambda$ is terminated with a complex load impedance as shown in the accompanying figure. Find the reflection coefficient at the load, the SWR on the line, the reflection coefficient at the input of the line, and the input impedance to the line.



2.9 A 75 Ω coaxial transmission line has a length of 2.0 cm and is terminated with a load impedance of $37.5 + j75\ \Omega$. If the relative permittivity of the line is 2.56 and the frequency is 3.0 GHz, find the input impedance to the line, the reflection coefficient at the load, the reflection coefficient at the input, and the SWR on the line.

2.11 A 100 Ω transmission line has an effective dielectric constant of 1.65. Find the shortest open-circuited length of this line that appears at its input as a capacitor of 5 pF at 2.5 GHz. Repeat for an inductance of 5 nH.

2.14 A radio transmitter is connected to an antenna having an impedance $80 + j40\ \Omega$ with a $50\ \Omega$ coaxial cable. If the $50\ \Omega$ transmitter can deliver 30 W when connected to a $50\ \Omega$ load, how much power is delivered to the antenna?

2.16 The transmission line circuit in the accompanying figure has $V_g = 15$ V rms, $Z_g = 75\ \Omega$, $Z_0 = 75\ \Omega$, $Z_L = 60 - j40\ \Omega$, and $\ell = 0.7\lambda$. Compute the power delivered to the load using three different techniques:

(a) Find Γ and compute

$$P_L = \left(\frac{V_g}{2}\right)^2 \frac{1}{Z_0}(1 - |\Gamma|^2);$$

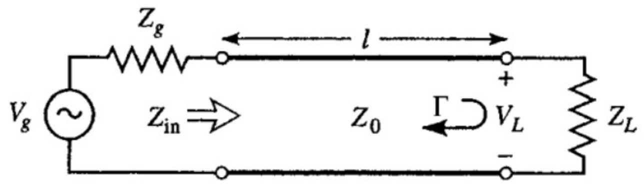
(b) find Z_{in} and compute

$$P_L = \left|\frac{V_g}{Z_g + Z_{\text{in}}}\right|^2 \text{Re}\{Z_{\text{in}}\};$$

(c) find V_L and compute

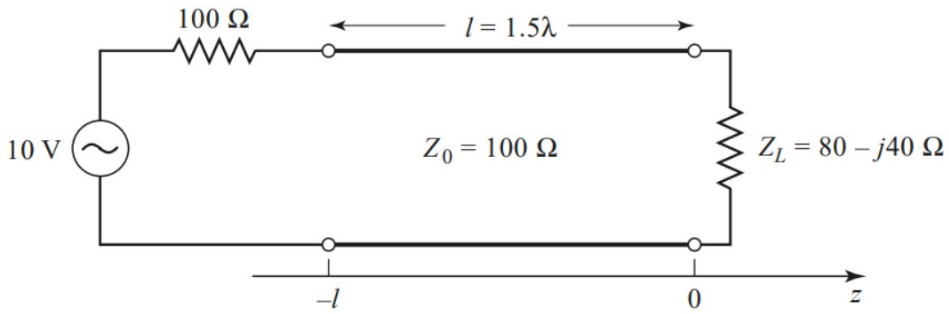
$$P_L = \left|\frac{V_L}{Z_L}\right|^2 \text{Re}\{Z_L\}.$$

Discuss the rationale for each of these methods. Which of these methods can be used if the line is not lossless?



2.17 For a purely reactive load impedance of the form $Z_L = jX$, show that the reflection coefficient magnitude $|\Gamma|$ is always unity. Assume that the characteristic impedance Z_0 is real.

2.19 A generator is connected to a transmission line as shown in the accompanying figure. Find the voltage as a function of z along the transmission line. Plot the magnitude of this voltage for $-\ell \leq z \leq 0$.



2.24 Design a quarter-wave matching transformer to match a 40 Ω load to a 75 Ω line. Plot the SWR for $0.5 \leq f/f_o \leq 2.0$, where f_o is the frequency at which the line is $\lambda/4$ long.

E1) A coaxial cable used for cable TV has a characteristic impedance of 75 Ω. The cable is filled with polyethylene, which has a relative permittivity of 2.25 and a loss tangent of 0.0004. Find the parameters (L, C). Also, find the parameter G at 1.0 GHz. If the manufacturer says that the attenuation is 0.204 dB/m at 1 GHz, find the value of R at this frequency. Hint: you can solve for R using "trial and error", searching for the value of R that will give you the correct attenuation.

E2) For the same cable as in Prob. E1, find the parameters (L, C, G, R) at 10 GHz. Hint: From the formulas for G and R, note how they vary with frequency, so you can use your results from Prob. E1.

E3) For the same cable as in Prob. E2, find the attenuation in dB/m at 10 GHz.